

Michael Rebarchik

Research Associate | Atomistic Materials Modeling and Computational Catalysis
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EDUCATION

Ph.D., Chemical Engineering - University of Wisconsin - Madison 2023

Thesis: “*Heterogeneous Thermo- and Electro-Catalysis over Single-Atom Catalysts*”

Honors B.S., Chemical Engineering (Summa Cum Laude) - Oregon State University 2018

SUMMARY

Computational materials scientist with 7+ years of experience in atomistic modeling with density functional theory (DFT), multiscale modeling, machine-learned interatomic potentials, and high-throughput computational screening to investigate catalysis and materials science challenges. Expertise in DFT (VASP, CP2K, FHI-aims), Python, and high-performance computing (HPC) for investigating catalytic surfaces, reaction mechanisms, and structure-property relationships. Proven track record of interdisciplinary collaboration, peer-reviewed publication, and development of computational methodologies bridging atomistic simulations, multiscale modeling, and experimental measurements for materials discovery and catalyst design.

EXPERIENCE

Research Associate Jan. 2025 – Present

University of Wisconsin – Madison, Department of Chemical & Biological Engineering Madison, WI

- Lead computational catalysis projects developing reaction-condition-cognizant active-site models to investigate catalytic performance and stability under electro- and thermo-catalytic reactions
- Perform atomistic simulations using machine-learned interatomic potentials, global optimization algorithms, and high-throughput computational workflows to investigate adsorbate-induced structural transformations, active site evolution of nanoparticle catalysts under reaction conditions
- Develop automated workflows for data collection, model training/validation, and model-processing pipelines, ensuring reproducibility, traceability, and efficient use of high-performance computing resources
- Collaborate with multidisciplinary research teams to translate modeling insights into research priorities, experimental design strategies, and data-driven decision making
- Mentor graduate students and junior researchers in computational methods, data analysis, and scientific communication

Catalyst Measurement Technical Specialist Aug. 2023 – Dec. 2024

Cummins Emissions Solutions – Chemistry & Kinetics Stoughton, WI

- Designed, constructed, and validated automated reactors and control systems to characterize catalytic materials, integrating sensors, data acquisition, and feedback control
- Developed new reaction protocols and kinetic models integrating experimental measurements and first-principles calculations to predict catalyst degradation mechanisms for metal-exchanged zeolites
- Led continuous improvement projects to optimize system throughput and data quality, reducing manual intervention, experimental variability, and user error through automated data analysis and standardized workflows
- Authored technical reports and peer-reviewed publications to support data-driven decision-making around new and existing products
- Provided technical guidance and operational support to laboratory personnel to coordinate scheduling and project execution

Graduate Research Assistant Aug. 2018 – Aug. 2023

University of Wisconsin – Madison, Department of Chemical & Biological Engineering Madison, WI

- Applied multiscale modeling and high-performance computing to integrate atomistic simulations with experimental data for heterogeneous catalysis
- Developed novel multiscale modeling frameworks based on first principles-derived data that enable benchmarking against experimental measurements, improving predictive accuracy
- Combined atomistic modeling of reaction mechanisms, multiscale modeling, and reaction kinetics experiments to investigate biomass conversion over transition metal catalysts
- Collaborated in multiple cross-disciplinary projects across UW-Madison and other universities
- Managed and maintained local Beowulf HPC cluster (~100 CPU nodes)

ADDITIONAL INDUSTRIAL EXPERIENCE

Physical Scientist Technician

Jan. 2018 – Aug. 2018

Rocky Mountain Research Center – U.S. Forest Service

Missoula, MT

- Developed web interfaces and dashboards to improve data accessibility to field wind measurements for wildfire early-warning systems

Process Engineer – MECOP Intern

Jun. 2017 – Dec. 2017

Rogue Valley Microdevices

Medford, OR

- Designed process control and data acquisition systems to reduce variability and enhance product throughput

Process Engineer – MECOP Intern

Mar. 2016 – Sep. 2016

Carestream Health Inc.

White City, OR

- Optimized unit operations (solvent recovery, deionized water system) to improve efficiency and reliability in the manufacturing of photosensitive films

SKILLS

- **Computational Chemistry & Atomistic Modeling** – Density functional theory (DFT; VASP, CP2K, FHI-aims, Gaussian), machine-learned interatomic potentials (MLIPs; Allegro, NequIP, MACE), kinetic/microkinetic modeling, *ab initio* molecular dynamics (AIMD), molecular dynamics (LAMMPS)
- **High-Performance Computing (HPC)** – Linux/Unix, Bash, Docker, workflow automation, job scheduling (Slurm, PBS), cluster management, database management
- **Modeling & Data Science:** Python (NumPy, pandas, scikit-learn, matplotlib), MATLAB, high-throughput screening, predictive modeling, supervised learning, model training/validation, Bayesian optimization, uncertainty quantification, genetic algorithms, basin-hopping global optimization
- **Project Management & Leadership** – Project coordination and timeline management, mentoring and training junior researchers, cross-functional and multi-institutional collaboration, scientific communication, technical reporting
- **Reaction Engineering & Characterization:** Reactor design, reaction engineering, GC, FTIR, MS, BET, HPLC

HONORS & AFFILIATIONS

Six Sigma Yellow Belt Certified

Member, **American Chemical Society (ACS)** and **American Institute of Chemical Engineers (AIChE)**

Neonatal Family Research Advisory Council, UnityPoint Health Meriter (2024-present)

Kokes Travel Award, North American Catalysis Society (2022)

Bird-Stewart-Lightfoot Wisconsin Distinguished Graduate Fellowship, University of Wisconsin (2021)

Donald E. Baldovin Graduate Fellowship, University of Wisconsin (2018)

Academic Achievement Award, Oregon State University (2018)

TEACHING EXPERIENCE

Guest Lecturer (25% of classes)

Spring 2025

University of Wisconsin - Madison – Prof. Manos Mavrikakis

Course: Molecular Modeling of Heterogeneous Catalysis (CBE 562)

Graduate Teaching Assistant

Fall 2020

University of Wisconsin - Madison – Prof. Ross Swaney

Course: Process Dynamics and Control (CBE 470)

Graduate Teaching Assistant

Fall 2019

University of Wisconsin - Madison – Prof. Manos Mavrikakis

Course: Kinetics and Catalysis (CBE 735)

PUBLICATIONS

Google Scholar: <https://scholar.google.com/citations?user=S0hMrbEAAAAJ>

1. Y. Ko, J. Choi, P. Połczyński, **M. Rebarchik**, A. C. Foucher, H. Szabat, J. Jiang, D. A. Cullen, M. Mavrikakis, Y. Yang, P. Zelenay, “Hollow-Structured Ni-N-C Catalysts for Highly Selective CO₂ Electroreduction”, *Applied Catalysis B: Environment and Energy* 397, 126919, DOI: [10.1016/j.apcatb.2026.126919](https://doi.org/10.1016/j.apcatb.2026.126919) (2026)
2. **M. Rebarchik** and M. Mavrikakis, “Assessing metal nitrides and metal carbides as supports for thermally stable single-atom catalysts”, *Chemical Engineering Journal* 532, 173933, DOI: [10.1016/j.cej.2026.173933](https://doi.org/10.1016/j.cej.2026.173933) (2026)
3. J. McAlpin, **M. Rebarchik**, H. Tupkar, Z. Jiang, S. Alemdar, M. Gallmeyer, J. E. Dahl, R. M. K. Carlson, B. O. Tkachenko, H. Hofstetter, C. F. M. Clewett, P. R. Schreiner, W. S. Loo, M. Mavrikakis, M. A. Gebbie, “Diamondoid-templated lithium single-cation conduction in solid-state organic electrolytes”, *Journal of Materials Chemistry A* 13, 36341-36350, DOI: [10.1039/D5TA03373J](https://doi.org/10.1039/D5TA03373J) (2025)
4. M. W. Harer, V. Mak, D. C. Kaluarachchi, P. Condit, L. Megna, E. Kipper, T. Freisleben, **M. Rebarchik**, M. Spangler, E. McQuade, S. Schwingle, and B. Gillick, “Design and implementation of a Neonatal Family Research Advisory Council (NFRAC)”, *Journal of Perinatology*, DOI: [10.1038/s41372-025-02491-y](https://doi.org/10.1038/s41372-025-02491-y) (2025)
5. Y. Yang, J. Feijoo, M. Figuras-Valls, C. Chen, C. Shi, M. V. F. Guzman, Y. M. Maombi, S. Liu, P. Jain, V. Briega-Martos, Z. Peng, Y. Shan, G. Lee, **M. Rebarchik**, L. Xu, C. J. Pollock, J. Jin, N. E. Soland, C. Wang, M. B. Salmeron, Z. Chen, Y. Han, M. Mavrikakis, P. Yang, “Operando probing dynamic migration of copper carbonyl during electrocatalytic CO₂ reduction”, *Nature Catalysis* 8, 579-594, DOI: [10.1038/s41929-025-01359-w](https://doi.org/10.1038/s41929-025-01359-w) (2025)
6. **M. Rebarchik***, E. Smith*, H. Chang, J. A. Dumesic, and M. Mavrikakis, “Selective Hydrogenation of Furfural Acetone over a Cu Catalyst: Combined Theoretical and Experimental Study”, *ACS Catalysis* 15, 6662-6672, DOI: [10.1021/acscatal.4c07231](https://doi.org/10.1021/acscatal.4c07231) (2025).
7. L. Xu*, **M. Rebarchik***, S. Bhandari, and M. Mavrikakis, “Adsorbate-Induced Adatom Formation on Au-Cu Bimetallic Alloys and Its Possible Consequences for CO₂ Electroreduction”, *Surface Science* 751, 122613, DOI: [10.1016/j.susc.2024.122613](https://doi.org/10.1016/j.susc.2024.122613) (2025)
8. D. Kiani, **M. Rebarchik**, T. Ryu, E. A. Lebrón-Rodríguez, W.-S. Lo, N. Ottinger, Y. Xi, Z. G. Liu, and I. Hermans, “Experimental Perspective on Occluded Cu_xO_y Nanoclusters in Hydrothermally Aged Cu-SSZ-13 SCR Catalysts”, *Chem Catalysis* 4 (6), 101012, DOI: [10.1016/j.cheecat.2024.101012](https://doi.org/10.1016/j.cheecat.2024.101012) (2024)
9. W. Xu*, R. Zeng*, **M. Rebarchik***, A. Posada-Borbón, H. Li, C. J. Pollock, M. Mavrikakis, and H. D. Abruña, “Atomically Dispersed Zn/Co-N-C as ORR Electrocatalysts for Alkaline Fuel Cells”, *Journal of the American Chemical Society* 146, 2593-2603, DOI: [10.1021/jacs.3c11355](https://doi.org/10.1021/jacs.3c11355) (2024).
10. **M. Rebarchik**, S. Bhandari, T. Kropp, and M. Mavrikakis, “Insights into the Oxygen Evolution Reaction on Graphene-Based Single-Atom Catalysts from First-Principles-Informed Microkinetic Modeling”, *ACS Catalysis* 13, 5225-5235, DOI: [10.1021/acscatal.3c00474](https://doi.org/10.1021/acscatal.3c00474) (2023).
11. **M. Rebarchik***, S. Bhandari*, T. Kropp, and M. Mavrikakis “How Noninnocent Spectator Species Improve the Oxygen Reduction Activity of Single-Atom Catalysts: Microkinetic Models from First-Principles Calculations”, *ACS Catalysis* 10, 9129-9135, DOI: [10.1021/acscatal.0c01642](https://doi.org/10.1021/acscatal.0c01642) (2020).
12. J. Ma, **M. Rebarchik**, S. Bhandari, G. W. Huber, M. Mavrikakis, and V. Zavala, “Exploiting Electricity Market Dynamics using Flexible Electrolysis Units for Retrofitting Methanol Synthesis”, *Energy & Environmental Science* 16, 2346-2357, DOI: [10.1039/D3EE00568B](https://doi.org/10.1039/D3EE00568B) (2023)
13. E. Gilcher, H. Chang, **M. Rebarchik**, G. W. Huber, and J. A. Dumesic, “Effects of Water Addition to Isopropanol for Hydrogenation of Compounds Derived from t-hydroxymethyl furfural over Pd, Ru, and Cu Catalysts”, *ACS Catalysis* 12, 10186-10198 DOI: [10.1021/acscatal.2c02445](https://doi.org/10.1021/acscatal.2c02445) (2022)
14. T. E. Kropp, **M. Rebarchik**, and M. Mavrikakis, “On the Active Site for Electrocatalytic Water Splitting on Late Transition Metals Embedded in Graphene”, *Catalysis Science & Technology* 9, 6793-6799, DOI: [10.1039/C9CY02006C](https://doi.org/10.1039/C9CY02006C) (2019).

*Authors contributed equally

PRESENTATIONS

1. **M. Rebarchik**, M. Mavrikakis, and J. A. Dumesic, “Hydrogenation Kinetics of Biomass-Derived Monomers over Transition Metal Catalysts: Combined First-Principles and Experimental Study” *2022 AIChE Annual Meeting*, Phoenix, AZ (Nov. 2022)
2. **M. Rebarchik** and M. Mavrikakis, “At Atomic-Scale View for Electrocatalysis.” *2022 AIChE Annual Meeting*, Phoenix, AZ (Nov. 2022)
3. **M. Rebarchik** and M. Mavrikakis, “Formic Acid Electro-Oxidation over Graphene-Based Single-Atom Catalysts: A Case Study on Coverage Effects” *2022 AIChE Annual Meeting*, Phoenix, AZ (Nov. 2022)
4. **M. Rebarchik**, S. Bhandari, T. E. Kropp, and M. Mavrikakis, “First-Principles Informed Microkinetic Modeling of Electrocatalytic Reactions on Graphene-Based Single-Atom Catalysts” *The 27th North American Catalysis Society Meeting*, New York, NY (May 2022)